

Advanced International Macroeconomics B (EI136)

Class 1

Patterns of international financial integration

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February 24, 2026

Outline of today class

- Aim and logistics of the course.
- International financial integration.
 - Definitions and patterns (Milesi-Ferretti 2025, 26).
 - Residency vs. nationality (Coppola et al. 2021).
 - Valuation effects, the U.S. “exorbitant privilege” (Bertaut et al. 2024).
 - Global imbalances and drivers of current accounts (Chinn and Ito 2025).

Aim and logistics of the course

- Overview of ongoing research in international macroeconomics.
- Lectures covering the topics in the syllabus, with the slides posted on the Moodle page before the course.
 - Focus on a few papers, briefly mention angle of additional papers.
- Grading and evaluation (templates on the moodle page):
 - Presentation of a paper from the reading list (30% of grade), conference format of 25-30 minutes, followed by class-wide discussion. Propose topics by April 3.
 - General participation in the class-wide discussion (10% of grade).
 - Referee report of a paper from the syllabus (30% of grade), about 3 pages, due by May 30.
 - Short literature review (30% of grade), 5-6 pages, due by May 30. Can be the basis of the literature review of your MPT or one of the chapters. Different segment of the reading list than the presentation and referee report.

Calendar

- Need to reschedule classes to avoid overlap with Gerzensee weeks.
- No class on February 17, March 10, March 31, April 28, and May 19.
- Replaced by classes on Monday (room S10, lever 4 petal 3) on March 2, March 23, April 13, May 11.

Week of	Monday	Tuesday
Feb 16 -20	----- Gerzensee exams -----	
Feb 23 -27		EI 136 CI 1 10-12, S1
March 2 - 6	EI 136 CI 2, 14-16, S10	EI 136 CI 3 10-12, S1
March 9 - 13	----- Gerzensee Macroeconomics 3 -----	
March 16 - 20		EI 136 CI 4 10-12, S1
March 23 - 27	EI 136 CI 5, 14-16, S10	EI 136 CI 6 10-12, S1
Mar 30 - Apr 3	----- Gerzensee Econometrics 3 -----	
Apr 6 - 10	----- Easter -----	
Apr 13 - 17	EI 136 CI 7, 14-16, S10	EI 136 CI 8 10-12, S1
Apr 20 - 24		EI 136 CI 9 10-12, S1
Apr 27 - May 1	----- Gerzensee Macroeconomics 4 -----	
May 4 - 8		EI 136 CI 10 10-12, S1
May 11 - 15	EI 136 CI 11, 14-16, S10	EI 136 CI 12 10-12, S1
May 18 - 22	----- Gerzensee Microeconomics 3 -----	
May 25 - 29		EI 136 CI 13 10-12, S1

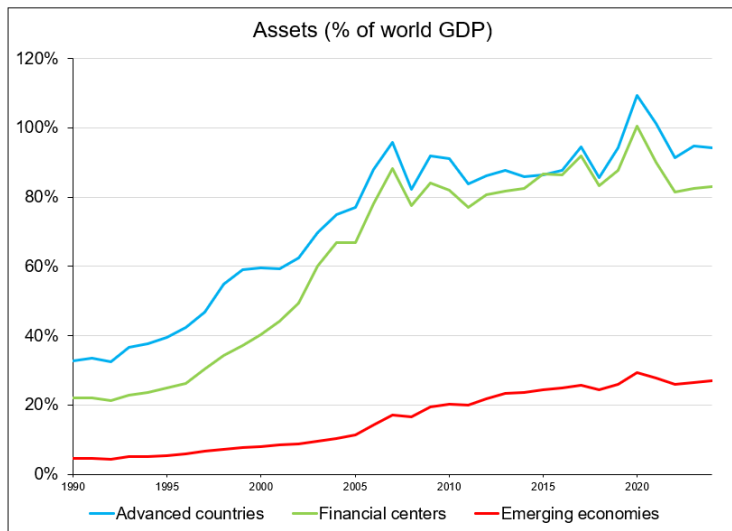
FINANCIAL GLOBALISATION

What is international financial integration?

- Value of countries' cross-border assets and liabilities (scaled by GDP).
 - *Net* assets (Asset - Liabilities) reflect saver / borrower position, *Gross* assets (Asset + Liabilities) reflect integration.
- Sharp increase since the mid-1990's until the crisis, especially in advanced economies (Milesi-Ferretti 2025, data available on Brookings website).
 - Decrease in 2022 do to a) lower asset prices, b) lower capital flows, c) higher nominal GDP(inflation), reducing IFI as % of GDP.
 - Growing role of financial centers (UK, Switzerland, Netherlands, Luxembourg, Singapore, small islands).
 - Large FDI assets (multinationals' special purpose entities), and portfolio equity (claims on mutual funds).
- A “gross” story more than a “net” story: “two-ways” financial integration where gross positions (and flows) dwarf net positions (and flows).

Some data: global IFI

- Rising, and heterogeneous since the 1990's, with large role of financial centers.



- Financial centers are mostly present in FDI, equity, and (a bit less) banking (Pogliani and Wooldridge 2022).

Figure 4. World FDI Abroad

Percent of world GDP

■ Financial centers
■ Advanced economies excl. financial centers
■ Emerging and developing economies

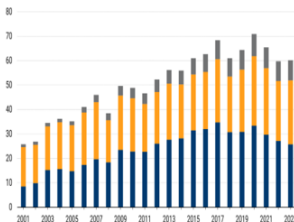


Figure 5. World Portfolio Equity Assets

Percent of world GDP

■ Financial centers
■ Advanced economies excl. financial centers
■ Emerging and developing economies

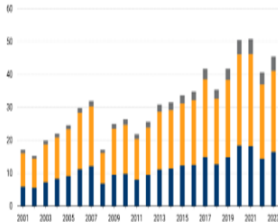


Figure 6. World Portfolio Debt Assets

Percent of world GDP

■ Financial centers
■ Advanced economies excl. financial centers
■ Emerging and developing economies

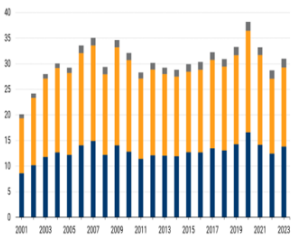
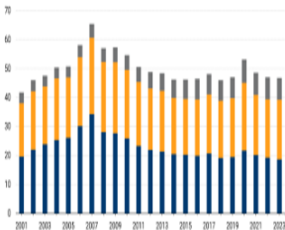


Figure 7. World Other Investment Assets

Percent of world GDP

■ Financial centers
■ Advanced economies excl. financial centers
■ Emerging and developing economies



Milesi-Ferretti, Gian Maria (2025). "External Wealth of Nations complete update, 2023"
Brookings

- Think of a German multinational owning an affiliate in Switzerland, which focuses on financial management. The Swiss affiliate owns a factory in France.
- *Residencyview*: an entity located in Switzerland is Swiss. Approach of the balance of payments.
 - Germany has an asset on Switzerland (the German affiliate is a Swiss resident).
 - Switzerland has an asset on France (the factory).
- *Nationalityview*: trace an affiliate back to its ultimate owner.
 - Germany has an asset on France (the factory). The fact that investment goes through Switzerland is irrelevant.
 - Switzerland has not asset and liability.

Nationality via securities holdings

- Residency - nationality gap largely driven by routing of investment through financial centers (Coppola et al. 2021).
- Data on securities (bonds and equity) issued by affiliates in financial centers, linked to nationality of parent company (where the investment goes) and nationality of investors (data on investment funds).
 - Complex: combine several different data sources.
- The correction increase advanced economies' holdings in emerging markets, especially BRIC's.
- FDI may not be as stable as usually thought: it is ultimately truly portfolio investment (residency treats holdings from the financial centers in the destination as FDI).
- Net position can be distorted (China).

- Re-allocation of US investors' bond holdings.

TABLE II
ESTIMATED NATIONALITY-BASED OUTWARD U.S. CORPORATE DEBT PORTFOLIOS, 2017

Destination	ISO code	TIC	With Reallocation	
			Position	Δ
Panel A: Selected non-tax haven countries				
Argentina	ARG	5	5	0
Australia	AUS	144	144	0
Brazil	BRA	8	50	42
Canada	CAN	390	393	4
China	CHN	3	48	45
France	FRA	118	120	2
Germany	DEU	60	80	20
India	IND	6	10	4
Indonesia	IDN	5	9	4
Italy	ITA	16	29	13
Japan	JPN	80	95	15
Mexico	MEX	58	58	0
Russia	RUS	0	12	12
Saudi Arabia	SAU	1	1	0
Spain	ESP	16	19	2
South Africa	ZAF	1	6	5
South Korea	KOR	11	11	0
Turkey	TUR	4	4	0
United Kingdom	GBR	308	326	18
Panel B: Selected tax havens				
Bermuda	BMU	30	0	-30
Cayman Islands	CYM	80	1	-79
Curaçao	CUW	3	0	-3
Guernsey	GGY	13	0	-13
Hong Kong	HKG	8	7	-1
Ireland	IRL	63	25	-39
Jersey	JEY	14	0	-14
Luxembourg	LUX	72	3	-69
Netherlands	NLD	179	95	-84
Panama	PAN	3	0	-3
British Virgin Islands	VGB	14	0	-14
Panel C: Total foreign corporate bond holdings				
Total foreign holdings	—	2,058	1,949	-109
Panel D: Domestic reallocation				
United States	USA	5,247*	5,356	109

Coppola, Antonia, Matteo Maggiori, Brent Neiman, and Jesse Schreger (2021). "Redrawing the Map of Global Capital Flows: The Role of Cross-Border Financing and Tax Havens", *Quarterly Journal of Economics* 136 (3), pp. 1499–1556.

- Re-allocation of holdings through financial centers.

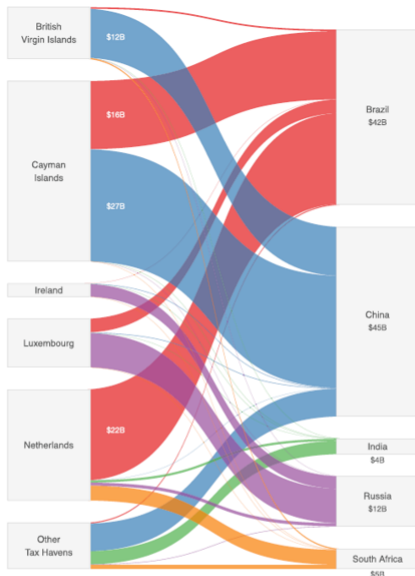


FIGURE II

Patterns of U.S. Corporate Bond Investment in Large Emerging Markets via Tax Havens, 2017

This figure shows the network through which companies in Brazil, China, India, Russia, and South Africa raise bond financing from U.S. investors via affiliates located in tax havens. The amounts on the right show the total amount of investment reallocated to each country from tax havens, and each colored line breaks down these amounts according to their source tax haven.

Coppola, Antonia, Matteo Maggiori, Brent Neiman, and Jesse Schreger (2021). "Redrawing the Map of Global Capital Flows: The Role of Cross-Border Financing and Tax Havens", *Quarterly Journal of Economics* 136 (3), pp. 1499–1556.

STOCK — FLOWS LINK : VALUATION

What drives assets and liabilities?

- Usual view: the dynamics of external assets and liabilities reflect capital flows (the current account).
- The dynamics of gross positions reflect 1) financial flows (trade and earnings on assets) and 2) capital gains on assets and liabilities..
- *Flows* are financial flows, i is the yield (dividends or interest), kg is the rate of capital gains, r is the total return: $i + kg$:

$$\begin{aligned}A_t &= A_{t-1} + Flows_t^A + kg_t^A A_{t-1} \\&= A_{t-1} + exports_t + \left(i_t^A + kg_t^A\right) A_{t-1} \\&= A_{t-1} + exports_t + r_t^A A_{t-1} \\L_t &= L_{t-1} + Flows_t^L + kg_t^L L_{t-1} \\&= L_{t-1} + imports_t + \left(i_t^L + kg_t^L\right) L_{t-1} \\&= L_{t-1} + imports_t + r_t^L L_{t-1}\end{aligned}$$

In net terms

- Net concepts: Net International Investment Position $N = A - L$, current account $CA = Flows^A - Flows^L$, trade balance $TB = exports - imports_t$.
- Large gross holdings magnify different returns between assets and liabilities:

$$\begin{aligned} N_t &= N_{t-1} + CA_t + \frac{kg_t^A + kg_t^L}{2} N_{t-1} + \left(kg_t^A - kg_t^L \right) \frac{A_{t-1} + L_{t-1}}{2} \\ &= N_{t-1} + TB_t + \frac{r_t^A + r_t^L}{2} N_{t-1} + \left(r_t^A - r_t^L \right) \frac{A_{t-1} + L_{t-1}}{2} \end{aligned}$$

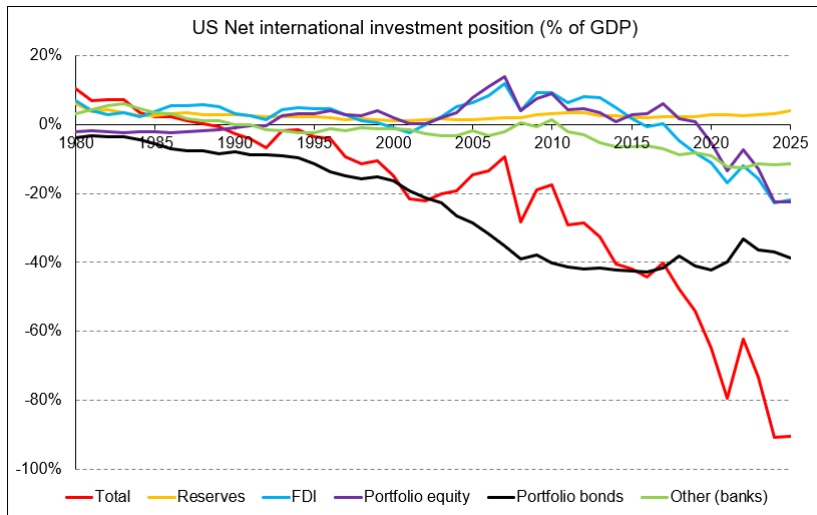
- Capital gains drive a large wedge between stock dynamics and flows. A model with only one asset has nothing to say about this.
- Example of different rates of returns: exchange rate, with assets in foreign currencies and liabilities in domestic currency.

Illustration with the U.S.

- Large increase in gross positions.
- Substantial heterogeneity.
 - Net creditor position in risky assets until recently (FDI and equity), net debtor position in non-contingent assets (bonds).
 - Liabilities in dollar, sizable share of assets (FDI and equity) in foreign currencies.
- Pre-crisis: net position is better than what capital flows imply. Capital gains reflect asset prices, exchange rates, and “other”, which is systematically in favor of the U.S.
 - U.S. earns larger capital gains on its assets than it pays on its liabilities.
 - Asset prices and exchange rate gains offset the current account deficit
- Post-crisis: net position deteriorated much faster than the current account deficit.
 - “World insurer” with price of risky assets falling in crisis times.
 - Rising monopoly profit margins in US raise US stock prices. Gain to foreign owners of US stocks, at the expense of US households (Atkeson and Perri 2025).

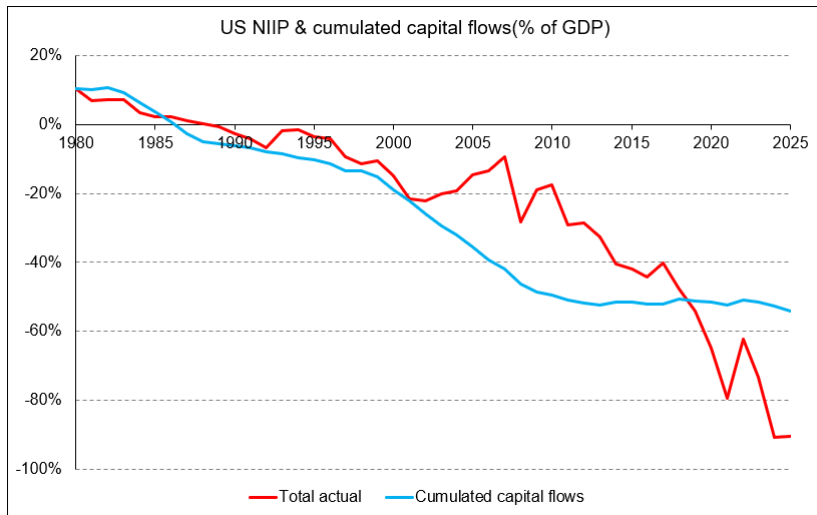
Changing net position

- US was long FDI and equity, but not any more.



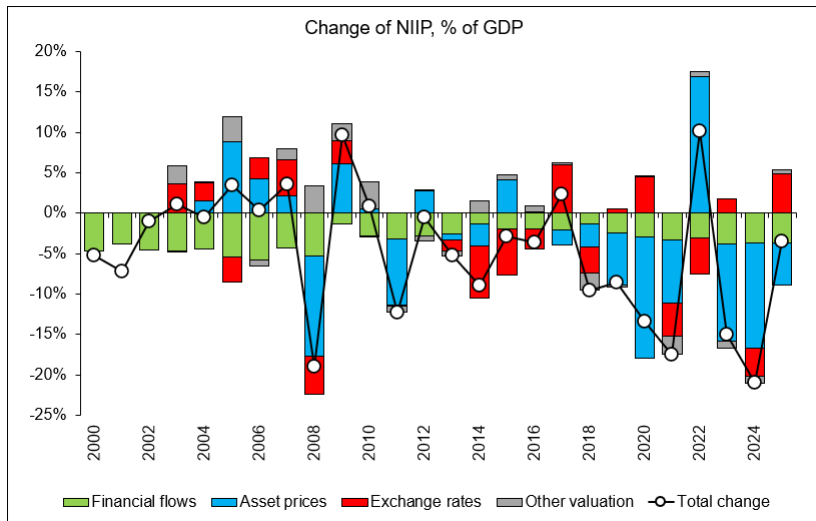
Net position vs. cumulated flows

- Large and persistent gaps.



Decomposing the NIIP dynamics

- Valuation effects of exchange rates and asset prices.



The US privilege

- Until 2018, capital gains were in the favor of the U.S., despite it being a debtor. The U.S. also had a surplus on its primary income (dividends + interest).
- The country got a higher rate of return on assets than it paid on liabilities.
- But is it statistically significant? Hard to asset due to volatility of returns (capital gains and losses), (Curcuru, Thomas and Warnock 2013).
 - Early work estimating long time series found a gap, subsequent work focusing on categories of investment (portfolio) with more reliable data found little gap.
 - Subsequent work found some gap, but smaller than early estimates.

A granular analysis

- Analysis relying on granular security level data (portfolio investment) for 2005-2022, and correcting for nationality (Bertaut et al. 2024).
- Overall, higher return on US assets (claims) than liabilities: 6.4% vs 5.9%, i.e. a small 0.5% gap.
 - Positive gap during normal types: 10.2% vs. 6.8%.
 - Negative gap during crises (2008-9, 2010-12, 2020-22): 1.6% vs. 4.7%.
 - The U.S. as a global insurer: exorbitant privilege and exorbitant duty.
- In normal times, the gap reflect the long equity, short bonds position.
- In crisis times, the gap reflects higher returns on liabilities than on assets within categories (equity, bonds), mostly from more robust asset prices in the U.S.
 - No return advantage of US Treasury over AAA foreign bonds (but quantities are different).
- Concentration of equity holdings in superstar firms for US liabilities, less so for US assets.

- A 3.3 percent favorable return gap in **normal times** from composition (long equity, short bonds), a 3.1 adverse gap in **crises** from intra-categories returns.

	Crisis	Non-crisis	Total
Equity claims	1.2	12.4	7.4
Equity liabilities	6.6	13.2	10.3
Bond claims	3.9	4.0	3.9
High grade sovereign	1.8	2.2	2.0
Other	5.7	5.6	5.6
Bond liabilities	3.2	3.1	3.2
Treasury/agency	3.9	2.3	3.0
Corporate	2.5	4.6	3.7
Total claims	1.6	10.2	6.4
Total liabilities	4.7	6.8	5.9
Total differential	-3.1	3.3	0.5
Composition	0.0	3.5	1.9
Return	-3.1	-0.2	-1.4
Exchange rate contributions:			
Total claims	-0.6	-0.7	-0.7
Equity claims	-0.6	-0.9	-0.8
Bond claims	-0.3	-0.4	-0.3
Total liabilities	0.0	0.0	0.0
Bond liabilities	0.0	0.0	0.0

- No return advantage in equity, even in normal times.
No return advantage in AAA sovereign bonds (but U.S. can issue more).

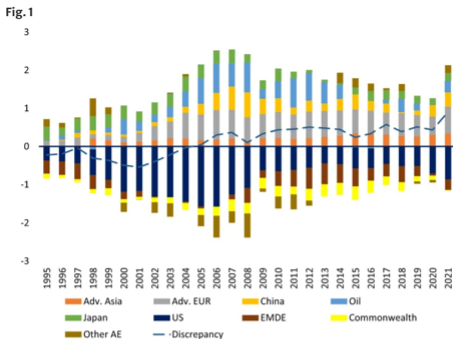
	Crisis	Non-crisis	Total
Equity claims	1.2	12.4	7.4
Equity liabilities	6.6	13.2	10.3
Bond claims	3.9	4.0	3.9
High grade sovereign	1.8	2.2	2.0
Other	5.7	5.6	5.6
Bond liabilities	3.2	3.1	3.2
Treasury/agency	3.9	2.3	3.0
Corporate	2.5	4.6	3.7
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Exchange rate contributions:			
Total claims	-0.6	-0.7	-0.7
Equity claims	-0.6	-0.9	-0.8
Bond claims	-0.3	-0.4	-0.3
Total liabilities	0.0	0.0	0.0
Bond liabilities	0.0	0.0	0.0

GLOBAL IMBALANCES

Imbalances in flows and stocks

- Before the 2008-9 crisis, current accounts across the world became much more dispersed, with some countries running large deficits and other large surpluses.
 - Different pattern from the 1980's, with large role of emerging economies as sources of funds.
- Particularly acute in the Euro-area with large surpluses in the North and deficits in the South.
- Imbalances in flows have narrowed after 2008, with lower total value of current account surpluses, and of deficits (Milesi-Ferretti, 2022, WEO 2025).
- Imbalances in stocks (NIIP) have increased.
 - Normal to some extent as even small flows add up to stocks, but mostly due to valuation changes.
 - Valuation losses in the U.S. only partially offset by valuation gains elsewhere.
 - Negative and growing stock discrepancy (unrecorded assets), mostly reflects portfolio equity (holdings through financial centers).

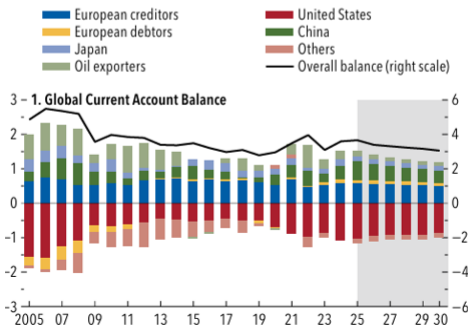
- Narrowing after the crisis. US is the main borrower, surpluses by oil producers and advanced Europe.



Source: Author's calculations based on IMF, Balance of Payments Statistics and World Economic Outlook

Global Current Account Balances (percent of world GDP). Note: See Section II for the definition of country groups.

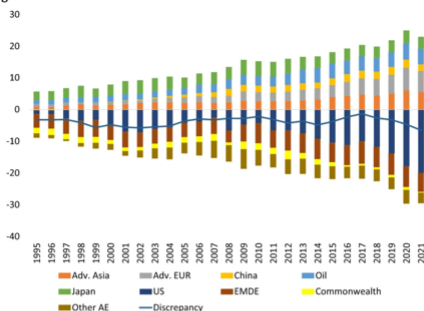
Milesi-Ferretti, Gian Maria (2023). "Many Creditors, One Large Debtor: Understanding the Buildup of Global Stock Imbalances After the Global Financial Crisis" *IMF Economic Review* 72, pp. 509-553.



IMF World Economic Outlook October 2025, chapter 1.

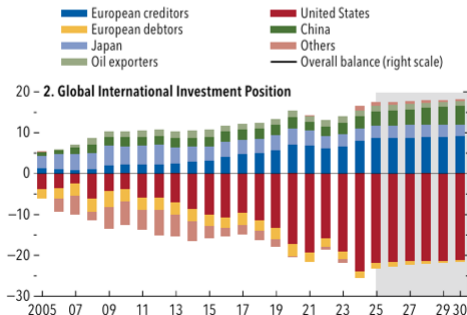
- Continued growth. US is biggest debtor. Rising creditors (advanced Asia and Europe, oil).

Fig. 2



Global Net International Investment Positions (percent of world GDP). Note: See Section II for the definition of country groups.

Milesi-Ferretti, Gian Maria (2023). "Many Creditors, One Large Debtor: Understanding the Buildup of Global Stock Imbalances After the Global Financial Crisis" *IMF Economic Review* 72, pp. 509-553



What drives current account?

- Long literature with macroeconomic (incl. fiscal balance), demographic (dependency ratio), and institutional (financial, legal development) factors.
- Recent event suggest a role for policy uncertainty, conflicts, and natural disasters (Chinn and Ito 2025).
 - Event analysis does not shows clearly significant patterns.
- Panel regressions on the current account, investment and savings (I focus on the first).
 - Role of twin deficits, demographics (sensitive), and institutional factors.
 - Global variables matters for emerging economies (Vix, US policy, recession).
 - Some effect of wars and disasters, but sensitive.
 - US policy uncertainty has a contrasted impact.

- Evidence of **twin deficits** (fiscal and current account). **Demographic factors** play some role, and **financial development and openness** end to lead to deficits.

Table 1a

Basic model augmented with saving glut variables, current account balance and nation:

	Current Account			
	(1) Full	(2) AE	(3) EMDE	(4) EME
Government budget balance	0.542 (0.060)***	0.364 (0.088)***	0.581 (0.081)***	0.273 (0.075)***
Net foreign assets (initial)	0.026 (0.005)***	0.016 (0.008)**	0.024 (0.006)***	0.035 (0.006)***
Relative income	0.104 (0.030)***	0.268 (0.072)***	0.095 (0.038)**	0.106 (0.039)***
Relative income squared	-0.032 (0.015)**	-0.101 (0.042)**	-0.029 (0.017)*	0.002 (0.029)
Dependency ratio (young)	-0.003 (0.010)	-0.048 (0.018)***	-0.005 (0.014)	-0.018 (0.013)
Dependency ratio (old)	0.001 (0.006)	-0.015 (0.014)	0.005 (0.008)	-0.035 (0.012)***
Financial Develop. (PCGDP)	-0.011 (0.007)	-0.019 (0.009)**	0.025 (0.027)	-0.050 (0.017)***
Legal development (LEGAL)	-0.002 (0.004)	-0.004 (0.006)	0.015 (0.010)	-0.016 (0.009)*
PCGDP × LEGAL	-0.006 (0.005)	-0.005 (0.011)	0.014 (0.011)	-0.028 (0.012)**
Financial open. (KAOPEN)	-0.005 (0.004)	-0.006 (0.005)	-0.013 (0.006)**	0.003 (0.007)
KAOPEN × LEGAL	0.002 (0.001)*	0.012 (0.002)***	-0.001 (0.002)	0.005 (0.002)**
KAOPEN × PCGDP	-0.008 (0.004)*	-0.010 (0.010)	-0.012 (0.006)*	-0.006 (0.007)
TOT volatility	0.116 (0.054)**	0.221 (0.152)	0.107 (0.055)*	0.202 (0.079)**
Avg. GDP growth	-0.066 (0.111)	-0.297 (0.166)*	-0.033 (0.124)	-0.048 (0.099)
Trade openness	-0.005 (0.006)	0.013 (0.007)*	-0.025 (0.009)***	-0.001 (0.012)
Oil exporting countries	0.015 (0.010)	-0.004 (0.019)	0.030 (0.011)***	0.025 (0.015)*
Observations	917	278	638	326
Adjusted R-squared	0.46	0.56	0.46	0.52
# of countries	126	35	91	42

Chinn, Menzie, and Hiro Ito (2025). "Current account dynamics and saving-investment nexus in a changing and uncertain world", *Journal of International Money and Finance* 151

Impact of global factors

- Emerging economies current account go in deficit in risk on times (lower VIX), or when global macroeconomic conditions tighten (higher US rate, recession).

Table 3
Impacts of global shocks – EMDE.

(a) Current Account					
	(1)	(2)	(3)	(4)	(5)
Max VIX in 5-yr period	0.044 (0.018)**				0.158 (0.038)***
Oil price		-0.016 (0.010)			-0.099 (0.024)***
US interest rate			0.045 (0.113)		-1.195 (0.336)***
Dummy for global recession				-0.005 (0.004)	-0.009 (0.005)**
N	552	585	585	585	552
Adj. R2	0.47	0.45	0.45	0.45	0.49

Chinn, Menzie, and Hiro Ito (2025). "Current account dynamics and saving-investment nexus in a changing and uncertain world", *Journal of International Money and Finance* 151

Wars and disasters

- Wars tend to generate a current account surplus (high savings, due to restrained consumption).
- Disaster tend to improve the current account, with parallel moves in savings and investment (higher after geophysical disasters).

Table 4
The impacts of disasters on CAB, NS, INV.

(a) Current Account						
	War (1)	Climato-logical (2)	Biological (3)	Geophysical (4)	ALL ex, bio. (5)	ALL (6)
# of war	0.004 (0.001)***				0.003 (0.001)**	0.003 (0.001)**
# of climatological disasters		0.001 (0.000)***			0.001 (0.000)***	0.001 (0.000)***
# of biological disasters			0.002 (0.001)**			0.001 (0.001)*
# of geophysical disasters				0.002 (0.000)***	0.000 (0.000)	0.000 (0.000)
N	647	647	647	647	647	647
Adj. R2	0.46	0.47	0.46	0.46	0.47	0.47

Chinn, Menzie, and Hiro Ito (2025). "Current account dynamics and saving-investment nexus in a changing and uncertain world", *Journal of International Money and Finance* 151

US policy uncertainty

- Higher uncertainty in US policy leads to deficits.
- Driven by fiscal and trade uncertainty, with opposite effect of monetary uncertainty.

Table 6

The impacts of U.S. uncertainty on CAB, NS, and INV.

(a) Current Account					
	(1)	(2)	(3)	(4)	(5)
US economic Policy Uncertainty	-0.038 (0.014)***				
US Monetary Policy Uncertainty		0.039 (0.017)**			0.043 (0.016)**
US Fiscal Policy Uncertainty			-0.015 (0.008)*		-0.015 (0.008)*
US Trade Policy Uncertainty				-0.004 (0.003)	-0.006 (0.003)**
N	552	552	552	552	552
Adj. R2	0.47	0.47	0.47	0.47	0.48

Chinn, Menzie, and Hiro Ito (2025). "Current account dynamics and saving-investment nexus in a changing and uncertain world", *Journal of International Money and Finance* 151

Do imbalances correct, and how?

- Consider how flows correct imbalances in stock of holdings (Alberola et al. 2020).
 - Creditor countries should have a current account deficit (trade or net asset income).
 - Regress the current account and its components on lagged net asset position.
- Highly heterogeneous pattern between creditors and debtors.
 - Debtors have trade surpluses, creditors do not have trade deficits.
 - Driven by trade balance asymmetry. Net asset income similarly destabilizing (surplus for creditors).
- Positive output gap leads to destabilizing dynamics for debtors.
- Imbalances can reflect that a country (say the US) is better able to produce safe liquid assets that are globally demanded (Benigno et al. 2025).
 - This shifts US output towards non-traded goods. If traded goods are the sector where innovation takes place, productivity and GDP growth slows (a financial resources “curse”).